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Quantum Mechanics
Lecture 4

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Recap

The concept and construct of wave packet (wave function) does address:

- Wave particle duality
- Uncertainty principle
- Dynamics of a particle

Schrödinger Equation for wave functions:

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi(x,t)}{\partial x^2}+V(x)\psi(x,t)=i\hbar\frac{\partial\psi(x,t)}{\partial t}$$

Classically,

$$\frac{p^2}{2m}+V=E$$

$$\underbrace{|\psi(x,t)|^2 \ \boxed{\text{Provides probability density}}}_{\text{Statistical Interpretation}} \rightarrow \underbrace{\int_{-\infty}^{+\infty}|\Psi(x,t)|^2dx=1}_{\text{Normalisation: A necessary condition for the wave functions to satisfy}}$$

Average/Expectation value

Since we can no longer speak with certainty about the position (momentum) of a particle, we can no longer guarantee the outcome of a single measurement of a physical quantity that depends on position (momentum).

- But, we can find the probable outcome of a meaurement i.e. can associate a probability of occurence of a value C_i upon measurement of the physical quantity C

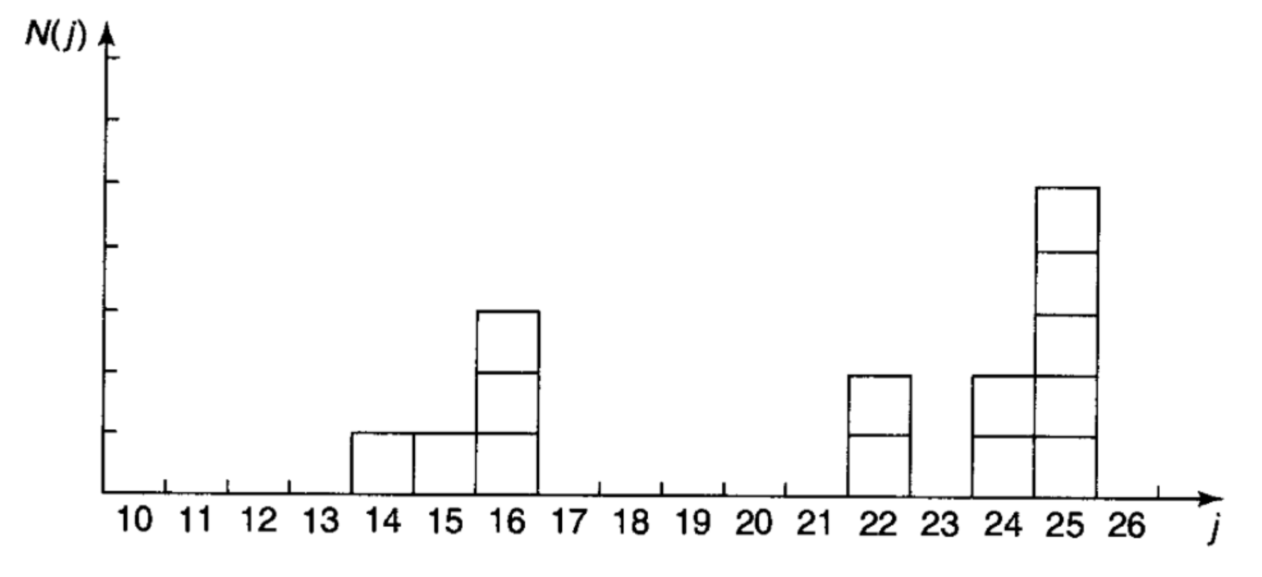
- When we perform a measurement on C N times($N \gg 1$), we find that the value C_i occurs n_i times. Then we can only talk about the average outcome of measurement on C which is given by

$$\langle C \rangle = \frac{1}{N}\sum_i n_i C_i, \quad N = \sum_i n_i$$

- Example: Imagine a room containing fourteen people, whose age are as follows:

one person aged 14
one person aged 15
three people aged 16
two people aged 22
two people aged 24
five people aged 25.

Consider, N(j) represents people with age `j'



Probability :

$$P(j) = \frac{N(j)}{N}.$$

$$\sum_{j=1}^{\infty} P(j) = 1.$$

So that it is legitimate!

Q: What is the average age ?

$$\frac{(14)+(15)+3(16)+2(22)+2(24)+5(25)}{14} = \frac{294}{14} = 21.$$

$$\langle j \rangle = \frac{\sum jN(j)}{N} = \sum_{j=0}^{\infty} jP(j).$$

Expectation value in Quantum Physics

Similarly for continuous variable, if probability density is given by $\rho(x)$

Probability: $$P_{ab} = \int_a^b \rho(x)\,dx,$$

Normalisation: $$\int_{-\infty}^{+\infty} \rho(x)\,dx = 1, \quad \equiv \quad \sum_{j=1}^{\infty} P(j) = 1.$$

Expectation value of position: $$\langle x \rangle = \int_{-\infty}^{+\infty} x\rho(x)\,dx, \quad \equiv \sum_j jP(j)$$

Expectation value of any function f(x) $$\langle f(x) \rangle = \int_{-\infty}^{+\infty} f(x)\rho(x)\,dx,$$

Given that $|\psi(x,t)|^2$ represents the probability of finding a particle between x to $x + \Delta x$,

Probability density

The expectation value of x for a particle in state ψ

$$\langle x \rangle = \int_{-\infty}^{+\infty} x|\Psi(x,t)|^2\,dx.$$

This is the most likely position of the particle, or average !

Expectation value is the average of repeated measurements on an ensemble of identically prepared systems (read same wave function), NOT the average of repeated measurements on one and the same system

Momentum Expectation value

Question: How do we find the expectation value of momentum ?

$$p = m\frac{dx}{dt} \rightarrow \langle p\rangle = m\frac{d\langle x\rangle}{dt}$$

$$\langle x\rangle = \int_{-\infty}^{+\infty} x|\Psi(x,t)|^2\,dx. \rightarrow \frac{d\langle x\rangle}{dt} = \int x\frac{\partial}{\partial t}|\Psi|^2 dx$$

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi(x,t)}{\partial x^2} + V(x)\psi(x,t) = i\hbar\frac{\partial\psi(x,t)}{\partial t} \rightarrow \boxed{-\frac{\hbar^2}{2m}\frac{\partial^2\psi^*(x,t)}{\partial x^2} + V(x)\psi^*(x,t) = -i\hbar\frac{\partial\psi^*(x,t)}{\partial t}}$$

$$\frac{\partial}{\partial t}|\psi|^2 = \psi^*\frac{\partial\psi}{\partial t} + \frac{\partial\psi^*}{\partial t}\psi \quad = \frac{i\hbar}{2m}\left(\psi^*\frac{\partial^2\psi}{\partial x^2} - \frac{\partial^2\psi^*}{\partial x^2}\psi\right) \quad = \frac{i\hbar}{2m}\frac{\partial}{\partial x}\left(\psi^*\frac{\partial\psi}{\partial x} - \frac{\partial\psi^*}{\partial x}\psi\right)$$

$$\frac{d\langle x\rangle}{dt} = \int x\frac{\partial}{\partial t}|\Psi|^2\,dx = \frac{i\hbar}{2m}\int x\frac{\partial}{\partial x}\left(\Psi^*\frac{\partial\Psi}{\partial x} - \frac{\partial\Psi^*}{\partial x}\Psi\right)dx.$$

Doing integration by parts and throwing the boundary terms:

$$\frac{d\langle x\rangle}{dt} = -\frac{i\hbar}{2m}\int\left(\Psi^*\frac{\partial\Psi}{\partial x} - \frac{\partial\Psi^*}{\partial x}\Psi\right)dx.$$

Performing another integration by parts:

$$\frac{d\langle x\rangle}{dt} = -\frac{i\hbar}{m}\int\Psi^*\frac{\partial\Psi}{\partial x}\,dx. \rightarrow \boxed{\langle p\rangle = m\frac{d\langle x\rangle}{dt} = -i\hbar\int\left(\Psi^*\frac{\partial\Psi}{\partial x}\right)dx.}$$

Concept of an operator

Expectation value of momentum:

$$\langle p \rangle = m\frac{d\langle x \rangle}{dt} = -i\hbar \int \left(\Psi^* \frac{\partial \Psi}{\partial x} \right) dx.$$

$$\langle p \rangle = \int \Psi^* (-i\hbar \frac{\partial}{\partial x}) \Psi dx \quad = \int \Psi^* \hat{p} \Psi dx$$

Replace momentum by:

$$\hat{p} \rightarrow -i\hbar \frac{\partial}{\partial x}$$

An operator !

An operator is an instruction to do something to the function that follows.

Position operator is a trivial one, that simply tells you to multiply by x

$$\langle x \rangle = \int_{-\infty}^{+\infty} x |\Psi(x,t)|^2 dx.$$

$$\langle x \rangle = \int \Psi^*(x) \Psi \, dx,$$

Therefore, for any variable which is a function of position and momenta

$$\langle Q(x,p) \rangle = \int \Psi^* Q(x, \frac{\hbar}{i}\frac{\partial}{\partial x}) \Psi \, dx.$$

Replace the appropriate powers of position and momenta in the function by the Operators.

Example: Consider, kinetic energy

$$T = \frac{1}{2}mv^2 = \frac{p^2}{2m},$$

$$\hat{p}^2 \rightarrow -\hbar^2 \frac{\partial^2}{\partial x^2}$$

$$\langle T \rangle = \frac{-\hbar^2}{2m} \int \Psi^* \frac{\partial^2 \Psi}{\partial x^2} dx.$$

Any operation of measurement requires an operator !

Commutation relation

Following the concept of an operator,

The commutation relation is defined as

$$[\hat{A}, \hat{B}] \equiv \hat{A}\hat{B} - \hat{B}\hat{A}$$

We must note that $\hat{A}\hat{B} \neq \hat{B}\hat{A}$

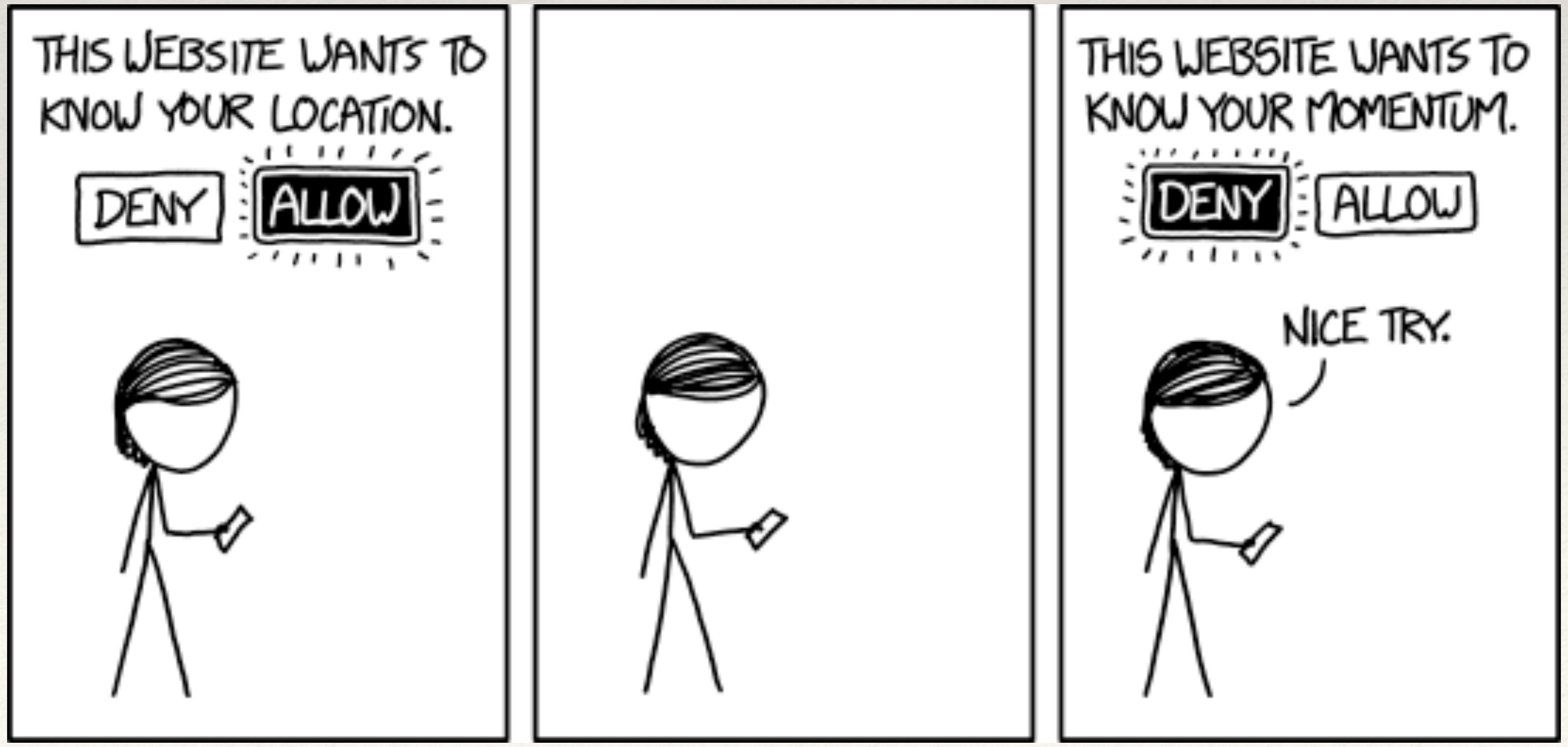
We wish to check the commutation relation of position and momentum

$$\hat{p} \rightarrow -i\hbar\frac{\partial}{\partial x}$$

$$[\hat{x}, \hat{p}]f(x) = x\frac{\hbar}{i}\frac{d}{dx}(f) - \frac{\hbar}{i}\frac{d}{dx}(xf) = \frac{\hbar}{i}\left[x\frac{df}{dx} - \left(f + x\frac{df}{dx}\right)\right] = i\hbar f,$$

$$\rightarrow \boxed{[\hat{x}, \hat{p}] = i\hbar.}$$

Everything stems from the uncertainty principle!



Momentum space wave function

Recall, from the definition of wave packet:

$$\psi(x,0) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dp\, \phi(p) e^{ipx/\hbar}$$

We multiply $\psi(x,0)$ by $e^{-ip'x/\hbar}$ and integrate all over x

$$\int_{-\infty}^{\infty} dx\, \psi(x,0) e^{-ip'x/\hbar} = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dx \int_{-\infty}^{\infty} dp\, \phi(p) e^{i(p-p')x/\hbar}$$

Using the fact that: $\int_{-\infty}^{\infty} dx\, e^{i(p-p')x/\hbar} = 2\pi\hbar \underbrace{\delta(p-p')}$

$\delta(p-p') = 0$, everywhere excepting at $p = p'$ ← Dirac Delta function

$$= \sqrt{2\pi\hbar} \int dp\ \phi(p)\ \delta(p-p')$$

$$= \sqrt{2\pi\hbar}\ \phi(p')$$

Fourier transform

$$\phi(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dx\, \psi(x,0) e^{-ipx/\hbar}$$

Let's check:

$$\int_{-\infty}^{\infty} dp\, \phi^*(p)\phi(p) = \int_{-\infty}^{\infty} dp\, \phi^*(p) \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dx\, \psi(x) e^{-ipx/\hbar}$$

$$= \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dx\, \psi(x) \int_{-\infty}^{\infty} dp\, \phi^*(p) e^{-ipx/\hbar}$$

$$= \int_{-\infty}^{\infty} dx\, \psi(x)\psi^*(x) = 1$$

$|\phi(p)|^2$ also depicts probability density to find a particle between $p \to p + dp$ in the momentum space.

$\phi(p)$ represents momentum space wave function.

Expectation values in momentum space

Let's check expectation value of momentum:

$$\langle p\rangle = \int_{-\infty}^{\infty} dx\, \psi^*(x)\frac{\hbar}{i}\frac{\partial\psi(x)}{\partial x} = \int_{-\infty}^{\infty} dx\, \psi^*(x)\frac{\hbar}{i}\frac{d}{dx}\frac{1}{\sqrt{2\pi\hbar}}\int_{-\infty}^{\infty} dp\, \phi(p)e^{ipx/\hbar}$$

$$= \int_{-\infty}^{\infty} dp\, \phi(p)p\frac{1}{\sqrt{2\pi\hbar}}\int_{-\infty}^{\infty} dx\, \psi^*(x)e^{ipx/\hbar}$$

$$= \int_{-\infty}^{\infty} dp\, \phi^*(p)p\phi(p)$$

Therefore, momentum is a trivial operator when it acts on momentum space wave function.

Like position is a trivial operator in position space

◆ Similarly, check:

$$\langle x\rangle = \int_{-\infty}^{\infty} dp\, \phi^*(p)\left(i\hbar\frac{\partial}{\partial p}\right)\phi(p)$$

$$\boxed{x = i\hbar\frac{\partial}{\partial p}}$$

Position is a non-trivial operator in momentum space

Standard deviation

Define: $\sigma_x = \sqrt{\langle x^2\rangle - \langle x\rangle^2}.$ $\equiv \Delta x$

This is the statistical definition of the uncertainty we have been talking about !

Similarly, $\sigma_p \equiv \Delta p = \sqrt{\langle p^2\rangle - \langle p\rangle^2}$

Given a wave function or mod square of that we can find all the expectation values and standard deviation!

Example: Consider the Gaussian distribution $\rho(x) = Ae^{-\lambda(x-a)^2}$, where A, a, and λ are constants. (a) Determine A, (b) Find <x>, <x²> , and σ, (c) Sketch the graph of $\rho(x)$.

Remember,

$$\rho(x) = |\Psi|^2$$

(a) $1 = \int_{-\infty}^{\infty} Ae^{-\lambda(x-a)^2}dx.$ Let $u \equiv x - a$, $du = dx$, $u : -\infty \to \infty$.

$$1 = A\int_{-\infty}^{\infty} e^{-\lambda u^2}du = A\sqrt{\frac{\pi}{\lambda}} \Rightarrow \boxed{A = \sqrt{\frac{\lambda}{\pi}}.}$$

(b)

$$\langle x\rangle = A\int_{-\infty}^{\infty} xe^{-\lambda(x-a)^2}dx = A\int_{-\infty}^{\infty}(u+a)e^{-\lambda u^2}du$$

$$= A\left[\int_{-\infty}^{\infty} ue^{-\lambda u^2}du + a\int_{-\infty}^{\infty} e^{-\lambda u^2}du\right] = A\left(0 + a\sqrt{\frac{\pi}{\lambda}}\right) = \boxed{a.}$$

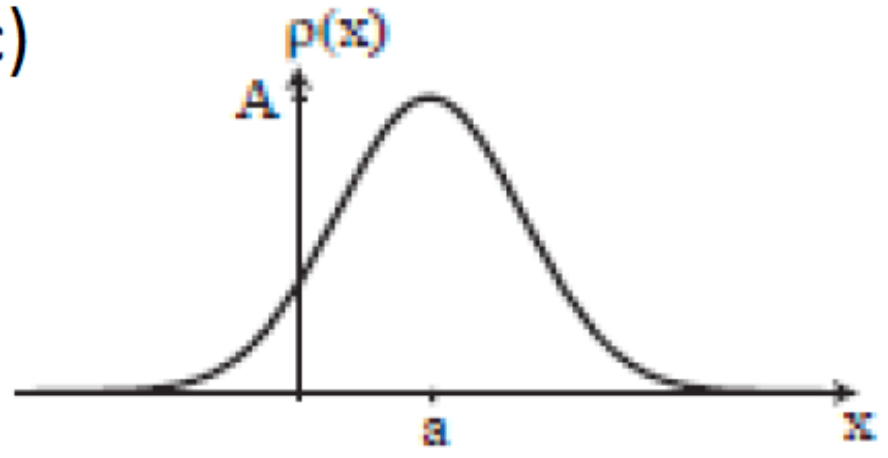
$$\langle x^2\rangle = A\int_{-\infty}^{\infty} x^2e^{-\lambda(x-a)^2}dx$$

$$= A\left\{\int_{-\infty}^{\infty} u^2e^{-\lambda u^2}du + 2a\int_{-\infty}^{\infty} ue^{-\lambda u^2}du + a^2\int_{-\infty}^{\infty} e^{-\lambda u^2}du\right\}$$

$$= A\left[\frac{1}{2\lambda}\sqrt{\frac{\pi}{\lambda}} + 0 + a^2\sqrt{\frac{\pi}{\lambda}}\right] = \boxed{a^2 + \frac{1}{2\lambda}.}$$

$$\sigma^2 = \langle x^2\rangle - \langle x\rangle^2 = a^2 + \frac{1}{2\lambda} - a^2 = \frac{1}{2\lambda}; \quad \boxed{\sigma = \frac{1}{\sqrt{2\lambda}}.}$$

(c)



Summary

- The outcome of any measurement in Quantum mechanics can only be specified by expectation value
- All the measurable quantities like position and momentum in quantum world are denoted by operators.
- The process of measurement is to act the respective operator on the wave function.
- The wave functions can be represented both in position and momentum space.
- Operators can take different forms in position and momentum space.